

Report

Ross McKenzie: Condensed Matter Physics, a Very Short Introduction

Diptanuj Sarkar (22MS038)
IISER Kolkata, 10.06.2026

1. Introduction

Condensed Matter Physics is a vast sub-field of Physics that primarily concerns itself with the study of matter in its “condensed” phases. The topics included in this field are diverse, ranging from abstract theoretical concepts to interdisciplinary applications in materials science, chemistry, and biology. All of these topics are unified by a few common concepts, and explored through a variety of simplified toy models that capture the essential physics of the phenomena that is being studied. In this report, I will be writing about my takeaways from Ross McKenzie’s book “Condensed Matter Physics: a Very Short Introduction” [1], which is meant to be a concise overview/introduction to the field of Condensed Matter Physics without much technical detail.

2. Condensed Matter Physics as a study of Emergence

All the materials around us have various properties that we can observe directly or measure using instruments. These properties are manifest on the scales of length, time, and energy that we are used to - **macroscopic**. Modern science is quite adept at measuring and characterizing these properties.

However, we are also aware that these materials are built from various atoms (which are, in turn, are made out of sub-atomic particles), drawn from a finite set of different types of atoms. Due to progress in our experimental capabilities, we have been able to determine that these constituent atoms (and sub-atomic particles) have their own set of rules that govern their behaviour. Due to Quantum Theory, a great deal is known about such **microscopic** behaviours - i.e. we are able to understand and characterize them well.

But this leaves us with a conundrum: How do we relate these microscopic properties of the constituents to the macroscopic properties of the materials that we observe?

A good example of this is discussed in Ross' book from the beginning - the differences between graphite and diamond. Both of these materials are composed of the very same type of atoms - carbon atoms. Microscopically, the carbon atoms themselves are identical in both materials, but the materials themselves can be easily observed to have very different properties. Graphite is soft and opaque, while diamond is hard and transparent. There are several other measured quantities that differ significantly in both of these materials. How do we explain these differences?

To begin with, we can note that the carbon atoms in graphite and diamond are arranged differently. In graphite, the carbon atoms are arranged in layers, with weak interactions between the layers. In diamond, the carbon atoms are arranged in a three-dimensional lattice, with strong interactions between all the atoms. From this it is easy to deduce that diamond is the harder material out of the two - the layers of graphite can simply be deformed as they are only weakly interacting with each other. Other properties, like color and conductivity of heat and electricity require more involved explanations, requiring quantum mechanics to explain the movement of electrons in the materials.

Both of these materials are examples of **states of matter**. Different states of matter have qualitatively different physical properties at *both* the microscopic and macroscopic levels. There is an unlimited number of distinct states of matter, and the sub-field of Condensed Matter Physics is concerned with characterizing and understanding these different states of matter.

Beyond this, however, Condensed Matter Physics is also defined by a particular approach to this study. In tune with other scientific disciplines, the study of Condensed Matter Physics is targeted towards:

1. Addressing fundamental questions

2. Finding unifying concepts and organizing principles

in analyzing various phenomena.

To that end, the central question of Condensed Matter Physics is as follows: **How do macroscopic properties emerge from microscopic properties?** In other words, how do the microscopic constituents of the material in question *interact* with each other to *collectively* give rise to macroscopic properties? Answering this question requires a determination of three things regarding any particular material:

1. Macroscopic Properties
2. Microscopic Properties
3. How the Macroscopic and Microscopic properties are connected

All of these lead to well-defined scientific questions, and it also makes it clear that the study of Condensed Matter Physics must be heavily reliant on diverse fields of specialization and progress in seemingly unrelated technologies to help us probe phenomena on vastly different scales of time, energy, and length.

2.1. Emergence

It is important at this point to compare Condensed Matter Physics with two other “fundamental” fields of Physics - Cosmology and Elementary Particle Physics. Both deal with phenomena on the opposite ends of the spectrum. Cosmology is concerned with the *large* scales, which means it deals with phenomena that take place over the age of the universe and on galactic (or larger) scales. On the opposite end, Elementary Particle Physics concerns itself with the extremely *small* scales, which means that it deals with the subatomic particles that constitute the universe. On the other hand, Condensed Matter Physics deals with the “complex”, i.e. systems that have many constituents interacting with each other.

This calls for the central idea of **emergence** - the fact that very often the whole is qualitatively different from the parts. Most macroscopic properties that we observe are irreducible to microscopic properties, and can only ever be explained via the *interactions* between a large number of microscopic constituents. There are of course some outliers, like the total mass or charge of the material in question, which is simply an additive phenomena that arises from the individual microscopic masses and charges of the constituents. Emergent phenomena very often present scientists with unexpected discoveries, great difficulty in making theoretical predictions, a qualitative difference between the microscopic and macroscopic properties, and universality - i.e. the fact that systems with different microscopic details can have the same macroscopic properties. This is in stark contrast to the more well known “reductionist” approach to science, which aims to develop a fundamental theory that can comprehensively describe all phenomena. It is clear that it takes original work to understand and predict

emergent phenomena starting from the microscopic facts - a sort of “building up” instead of a “building down” approach.

3. Symmetry, Order, and Landau Theory

To enable a study of the states of matter, we must first be able to identify them based on quantitative grounds. This is required, as an identification made purely on the grounds of qualitative difference can lead to disagreement of the definitions of various states of matter.

3.1. Phase transitions

To understand how we can approach such a quantitative method, we need to identify ways in which we can detect if a material has changed the state in which it is. A simple way of doing this is to measure the temperature of the substance as we cool or heat it. A transition in state is marked by a certain *abrupt* change in one or more of its properties - it is a “tipping point” phenomena. A small change of a fraction of a degree in temperature can bring about large changes in the properties of the material around a phase transition. In the words of Ross McKenzie, “Tipping points show that quantitative differences can become qualitative differences”.

If we plot the temperature of a fixed amount of material against time, while providing heat energy to it at a constant rate, we will observe certain plateaus every time it undergoes a phase transition. This is because the heat energy that we are providing to the material is being used up in changing the state of the material, instead of increasing its temperature. For example, when we heat ice (at normal atmospheric pressure), it will remain at 0 degrees Celsius until all of it has melted into liquid water. Once all of it has melted, the temperature will start rising again until it reaches 100 degrees Celsius, at which point it will again plateau until all of the water has evaporated into steam. If we change up the conditions of pressure during this experiment, the graph of Temperature versus time would remain qualitatively the same, but the exact temperatures at which the plateaus show up would change. This method is very frequently used to identify and discover new states of matter, by cooling materials to lower and lower temperatures. Superconductors and superfluids are some of the many states of matter that were discovered using this method.

The most common way of identifying states of matter *at equilibrium* is through the measurement of the external conditions at which the material is placed. For example, at room temperature and pressure, if we pour out a glass of water it will remain in that state (barring evaporation). If we instead place a block of ice in the same conditions, we can be certain that it will melt eventually and turn into liquid water. The same is true for a container of steam, which will eventually condensed and turn into liquid water. Thus, we can conclude that at equilibrium at room temperature and pressure, liquid water is the *stable phase* of the substance. We can prepare diagrams to represent the conditions under which a certain material exists in a particular state of matter, and these are called **phase diagrams**. This is an elegant solution to the problem of quantifying states of matter, as it does not require us to tabulate a long list of properties

of the states. Simply specifying the external conditions is enough to identify the state of matter that we are dealing with.

This method also has its downsides, as there exist states of matter that are *meta-stable* in conditions outside of their usual stable conditions. A good example is diamond, which has been determined to be the stable form of carbon at enormous temperature and pressure. Yet, we can observe diamonds in regular conditions. This is because diamond is a very long-lived metastable state of carbon in these conditions. If we could wait for millions of years for it to *equilibrate* to the external conditions, it would slowly turn into graphite.

3.2. Symmetry as a unifying principle

Note that while the methods described above can help us identify phases of a material, it does not provide any organization. As in any discipline of science, we require some unifying idea that will help us classify various states of matter. **Symmetry** is a concept that helps us undertake a classification of the states of matter.

A symmetry of an object is defined as a transformation that keeps it unchanged. These can be discrete, as in the three-fold rotational symmetry of an equilateral triangle, or continuous, as in the continuous rotational symmetry of a circle. In physics, the symmetries that a system respects dictate what is possible for the system. In the context of Condensed Matter Physics, this shows up most clearly in the case of crystals. Most minerals are found as crystals, and it was determined early on that crystals of the same mineral always has the same shape (or, symmetries). It was also discovered that most crystals seem to be made of smaller units that share the exact same shape as the larger piece.

These facts lead us to conclude that if matter in the form of crystals is made up of discrete units, these units need to have a shape that is able to *tile* the space that is occupied by the crystal. Only certain, finite number of such “units” exist in a given dimension - this may be determined using the idea of symmetries. For example, a unit with a fivefold rotational symmetry can not tile a 2 dimensional plane, whereas a square (with a 4-fold rotational symmetry) can. Once we have determined the unit or the motif of a crystal, we simply need to translate it in space to recreate the entire crystal. Bravais and Schoenflies determined mathematically that there are a total of 14 such units that can tile 3 dimensional space, and there are only 5 possible units in two dimensions. In this way, we can see that symmetries restrict what is possible.

After the discovery of X-Ray diffraction, it became possible to determine the microscopic structure of materials. Using this information, it became possible to not only classify states of matter but also explain certain macroscopic properties of the material. An example of this are snowflakes. Whenever a snowflake forms, it happens to form with a clear 6-fold rotational symmetry. Using diffraction techniques, it was

determined that the water molecules in the snowflake also freeze into a crystal that has 6-fold symmetry in its units. It is this microscopic symmetry that is manifested in the macroscopic snowflake.

Symmetry does not only constrain the type of crystal lattice, but it also determines several properties of the crystal. A simple example of this is the propagation of mechanical waves through a crystal. A crystal that is highly symmetric, like one with a cubic unit has two ways in which it can be deformed:

1. A compression along any axis, changing the side lengths of the cube but keeping the angles between the sides the same.
2. A shear along any face of the cube, changing the angles between faces but keeping the side-lengths the same.

Corresponding to the two possible deformations of the unit cell of the crystal, there are two different ways in which a mechanical wave can propagate through the crystal. One is longitudinal, corresponding to the compression, and the other is the transverse, corresponding to the shear. Thus, the macroscopic property of mechanical wave propagation is seen to arise directly from microscopic symmetries. Once a principle such as this one has been identified, we can use it either way: knowing the symmetries of the crystal we can determine its properties, or we can investigate the properties of the material to determine its underlying symmetries.

3.3. Order

The symmetries of a system determine the ordering of the constituents of the system, and an analysis of the ordering of the constituents of a system can lead us to an understanding of the symmetries of the system. It is natural to translate the fact represented by symmetries into the terminology of “order”, as order means patterns and most of science is based a search for patterns in the natural world. This is reflected in the Liquid Crystal phase of matter. Liquid Crystals are made out of cylindrical molecules, and in the liquid crystal phase they have orientational order but no positional order. In terms of symmetries, this means that the liquid crystal phase retains the continuous translational symmetry of the liquid phase, but it has lost the continuous rotational symmetry that the liquid possesses. In the case of ferromagnetic states, the magnetic moments of the constituent atoms align with each other to produce a net macroscopic magnetic moment. The ordering and symmetries of many other states of matter are not as clear, as in the case of superconductors, antiferromagnets, or superliquids.

With the organizing principles of order/symmetry in place, Lev Landau was able to propose a simple theory that could be used to describe the properties of a material in the vicinity of a phase transition. The theory centers around the so-called “order parameter”, which can be roughly described as a quantitative measure of the ordering in the system. This order parameter could be a scalar, like the average magnetisation

in the case of simple magnetic systems, or vectors with multiple components, as in the case of liquid crystals. Additionally, the order parameter would only have a non-zero value in an ordered state. Landau argued that a free energy functional of the material could be constructed as a simple function of the order parameter near the critical point of a phase transition. Such a free energy functional would have a form that would be constrained by the symmetries of the material. Since we assume that the system should always be in a state of minimum free energy, we can simply find the equilibrium value(s) of the order parameter by minimizing this function. The method is intrinsically simple, and it shows us that most of the microscopic details of the material is immaterial to the phase transition itself. This tells us that there is a certain “universality” to phase transitions.

An important concept that is associated with Landau’s Theory of phase transitions is **Spontaneous Symmetry Breaking**. Suppose we have matter in a state that has some sort of symmetry. Below the critical temperature of a phase transition, it may happen that there are many (finite, or infinite) states that have a lower free energy but also do not possess the same symmetry as the high temperature state. In such a case, if we allow the material to equilibrate it will relax to one of these lower energy states. This phenomena is called spontaneous symmetry breaking.

The idea of the order parameter also allows us to quantify the **rigidity** of a state of matter, as we can now define it to be the value of energy required to create a certain amount of change in the order parameter of the system in a particular state.

4. The effect of Dimensionality

While we reside in 3 dimensions, it is fruitful for theoretical purposes to consider the behaviour of systems in other dimensions. With advances in technology, it has become possible to create physical systems that are 2- or 1-dimensional in nature. This consideration of lower dimensions turned out to be very productive, as it has led to a deeper understanding of phase transitions in general and has shown us that dimensionality affects the nature of a phase transition.

4.1. The Ising model

Mathematical models for physical systems need to be constructed with a particular balance in mind - they need to be simple enough for numerics and mathematical analysis, but also complicated enough to capture all the relevant details of a physical phenomena. The most famous example of such a model is the Ising model. It was originally envisioned as a model for magnetism, but it has since found applications in a wide variety of fields.

The Ising Model is able to demonstrate all the crucial features of a phase transition, including spontaneous symmetry breaking, ordering, critical points, and abrupt changes in physical properties. The model consists of a lattice of sites, with a spin at each site that can take one of two values - up or down. Each spin interacts with its immediate neighbours (the number depends on the lattice that we are using), and the system incurs an energy penalty when neighbouring spins are aligned in opposite directions. At finite temperatures, the spins also demonstrate random fluctuations, and it is the competition between this tendency to randomly flip spins and the energy costs of interaction that drives the interesting behaviour of the system.

Originally, Ernst Ising was able to exactly solve the 1-dimensional Ising Model, which means that he was able to analytically determine the partition function and thermodynamic properties of the model for any temperature, without approximations. From this, he showed that it was impossible for the 1-dimensional Ising Model to undergo a phase transition, and he erroneously extended this idea to the 2-dimensional Ising Model as well. It was shown in 1944 by Lars Onsager that the 2-dimensional Ising Model on a square lattice does in fact undergo a phase transition. The solution for the 2-dimensional Ising Model is significantly more complicated than that of the 1-dimensional case, and is generally not expected to be solvable for more complicated lattices. What was notable with Onsager's solution was that its predictions did not match with Landau's Theory, and it established an important fact: short-range interactions can lead to long-range order. In this way, the 2-dimensional Ising Model exhibits rigidity - the ordering of spins is not as fragile as Landau might have us believe.

4.2. 2-dimensional systems in the laboratory

It has become possible to create systems where the relevant phenomena is restricted to only 2 dimensions. There are several ways to achieve this:

- A perfect crystal is cleaved along a face, which leaves a perfectly 2-dimensional surface. Now, material can be deposited onto this face in a thin film, and can exhibit 2-dimensional behavior. 2-dimensional superfluid was made using this method, by depositing a thin film of Helium-4 on top of a graphite face.
- Certain materials can be prepared layer-by-layer. By altering the chemical properties of adjacent layers, it is possible to create materials where electrons are confined to a 2-dimensional plane.
- There exist materials called quasi-two-dimensional. Here, the material is made up of 2-dimensional layers that only weakly interact with each other. A classic example of this is graphite, with only weak van der Waals forces acting between adjacent layers.

4.3. New phases of matter in 2 dimensions

In 3 dimensions, the atoms in a crystal lattice vibrate around their equilibrium positions due to thermal energy. On increasing temperature, these vibrations increase in magnitude until it reaches the lengthscales of the lattice spacing. At this point, the crystal is said to “melt”, and it transitions into a phase where it has no long-range order. However, in 2 dimensions, the situation is different. David Mermin showed in 1986 that perfect crystals can not exist in 2 dimensions - the thermal fluctuations are too strong for long-range order to exist. Due to this, a curious new phase called the *hexatic* phase arises - here the atoms have orientational order, but no positional order. This phase exists as an intermediate between the crystal at zero temperature, and the liquid at high temperature. This phase was predicted theoretically first, and it was later observed experimentally in the laboratory.

It is also important to note that this “melting” of the crystal in 2 dimensions is also qualitatively different from that in 3 dimensions. In 2 dimensions, there are topological defects which are present as pairs in the low temperature phases, and upon increase of temperature they become unbound from each other and destroy the long-range order. Due to this, transition to states with long-range order, such as superconductors, superfluids, or magnetic states, are also qualitatively different in 2 dimensions compared to 3 dimensions.

4.4. Quasi-two-dimensional systems

Motivated by a search for a high-temperature superconductor, several quasi-two-dimensional materials have been synthesized in recent years. These are usually compounds of copper and oxygen with several other atoms thrown in the mix. The repeat units of this material is very complex, and it has layers of copper and oxygen atoms that are separated from each other by other atoms. This makes the electrons on

the layers subject to a quasi-two-dimensionality, as it is far easier for them to move within a layer than between layers.

While these materials did demonstrate superconductivity at significantly higher temperatures than reached before, there are several interesting features about their phase diagrams. Apart from the Fermi Liquid (normal metal), Antiferromagnetic insulator, and superconductor phases, these materials also demonstrate two new phases - **Pseudogap** and **Strange Metal**. These are conducting states, and theories developed to explain the previous phases are unable to explain the behavior of these states.

In the pseudogap phase, the material conducts like a semiconductor, except in a few directions. In those directions, the electrons behave as if they are present in a metal. In the strange metal phase, the electrons move collectively in a fashion that makes the idea of isolated electrons meaningless. Fermi Liquid theory, which forms the basis of our understanding of normal metals, describes resistivity of the metal as something that arises from the scattering of well-defined quasiparticles at some rate. This rate is determined by the temperature, and at low enough temperatures the time-scales of this scattering are proportional to the square of the temperature. However, in the strange metal phase we can observe a linear relationship between the resistivity of the material and temperature [2].

All theoretical and experimental studies seem to point to the fact that these two phases are intrinsically two-dimensional. Thus, changing around the dimensionality of the system under investigation can lead to otherwise unexpected behaviours.

5. Criticality and Renormalization Group

In most disciplines of science, we usually seek to classify phenomena into distinct categories. But beyond such a classification, there are underlying *universal* principles that govern these phenomena. Similarly, in the context of Condensed Matter Physics, we can see that when a any material is driven to near a critical point of a phase transition, many of the microscopic details of the material become irrelevant to the behaviour of the material. This is what is meant by *universality*. This is demonstrated in Landau's Theory, as mentioned before, and it is typically seen in the form of **scaling laws** for various physical observables near a critical point. It turns out that it is possible to arrive at a precise classification of phase transition based on these scaling laws. These classes are called *universality classes*, and can have seemingly unrelated phase transitions grouped together. For example, superfluid, superconductor, and a subset of magnetic transitions fall into the same universality class.

The only details that are relevant to such a classification is the symmetry of the state, and the dimensionality of the system - nothing else seems to matter.

5.1. Universality near the liquid-gas transition

In 1945, Edward A. Guggenheim collected data of density versus temperature of various pure substances near their respective liquid-gas critical points. He then rescaled them with their values at the critical point. He found that the data for various different substances all collapsed onto a single curve. This shows that the values attained by the macroscopic observables of the substances are all, in some sense, “similar” near the critical point and are simple rescaled versions of each other.

5.2. Universality of Critical exponents

Let us look at one of the critical exponents used to quantify the behaviour of a system near a critical point - α . This is the critical exponent of specific heat, and is given by the equation $C = At^{-\alpha}$, where t is the reduced temperature, $t = \frac{T-T_c}{T_c}$, and T_c is the critical temperature. Surprisingly, a large class of magnetic substances have the exact same value of α near their magnetic transition, regardless of their chemical composition or crystal structure. Even superfluids have the same value of α near their phase transition. There are several other critical exponents, such as for the compressibility γ , order parameter ν , and some more.

What is suprising is that not all of these exponents are independent of each other, and that all materials in the same universality class share the exact same set of critical exponents.

Landau's theory produced the same set of critical exponents for all transitions, and even then the values were slightly off from those determined via experiment. This was

because Landau's Theory ignored thermal fluctuations - due to these the value of the order parameter is not the same at each point in space and time in the system.

5.3. Renormalization

Looking at a visual representation of a 2-dimensional Ising model, made out of black and white squares to denote up and down spins, we can see that certain types of scaling are demonstrated. Let us lay out our procedure for "zooming out" on the model. First, we select a lengthscale by specifying a block of some size, say 3×3 . Now, we simply replace the spins in any such 3×3 block with a single spin, whose color is determined by the color of the majority of the spins in that block. We then repeat this process, zooming out or blurring our vision of the model.

There are three outcomes of this procedure, if we continue to do it enough times:

1. Under the critical temperature: Here, we can see that large domains of a specific type of spin has formed. On carrying out the scaling operations, the entire system will take on one single spin, either all up or down. This is simply the spontaneous symmetry broken state of the system.
2. At Critical temperature: On applying the scaling procedure repeatedly, we can see that the system *looks the same* at any lengthscale - it is scale-free. What this means in our particular context is that we can always find a path through the entire system that passes through either black or white.
3. Above the critical temperature: Here, regardless of the lengthscale that we scale to, the system remains a random collection of black and white spins, with no long-range order.

The procedure was first developed by Leo Kadanoff in 1966, and later was formalized into a theory by Kenneth G. Wilson in the 1970s - known by the term **renormalization group**. Wilson essentially incorporated the effects of thermal fluctuations into Landau's theory, and the procedures prescribed by Wilson's theory led to a set of values for critical exponents for several universality groups that matched the experimentally determined values. More importantly, it tells us *how* universality emerges in such a diverse set of materials. As a material approaches criticality, its macroscopic properties are no longer determined by single atoms, but rather by very many atoms in *domains*. For systems in the same universality class, how these domains interact with each other is the same, which leads to universality. These interactions are determined simply by the symmetries and dimensionality of the system.

Generally speaking, this tells us that at each lengthscale there is a certain *effective theory* for a system, which describes the system *at that lengthscale* entirely. Wilson's methods give us a way to arrive at the effective theory at larger lengthscales, starting from the effective theory at the microscopic lengthscale. Often, as we use this proce-

dure to “coarse-grain” or “zoom out” on a system, the effective theory simplifies and becomes more amenable to mathematical analysis and numerics.

6. Macroscopic Quantum Matter

It is well known that the predictions that arrive from Quantum Theory are very non-intuitive. The prototypical example of this is Schrodinger's Cat, which is simultaneously alive and dead - in quantum superposition. Such a state is real, and this phenomena (and several others) that are predicted by quantum theory have been experimentally verified to incredible degrees of accuracy. However, we do not observe such strange phenomena in our day to day lives, since we live in the macroscopic world. It is generally held that quantum mechanics truly starts to be the dominant theory at very small, microscopic lengthscales.

However, Condensed Matter has several examples of states of matter where the boundary between the quantum and classical domains is shifted significantly. These are known as Macroscopic Quantum Phenomena. There are three major markers for such a phenomena:

1. Quantum discreteness: Quantum phenomena are different from classical phenomena in that the values attained by physical observables (energy, angular momentum, etc.) are quantized - they can only change in discrete steps. In Classical systems, these take values from a continuous set.
2. Wave-particle duality: In quantum mechanics, particles exhibit properties of both waves and particles. This can lead to interference and diffraction phenomena. Classically, particles cannot interfere with each other (or themselves) as they are localized in time and space.
3. Appearance of certain fundamental constants: Constants such as the Planck constant $\hbar$ and the elementary charge e take values that are typically thought of as microscopic. The appearance of such fundamental constants in expressions for macroscopic properties of materials is a sign of a Macroscopic Quantum Phenomenon.

In the following sections, I will describe various states of matter that demonstrate such macroscopic quantum phenomena in brief, and then go over the phenomena in question.

6.1. Superconductors

Discovered in 1911 by Kamerlingh Onnes in Mercury, a superconductor is a state of matter where electricity flows with zero resistance. In normal metals, electrons carrying the current scatter off of imperfections in the material, and off of the lattice vibrations, causing a dissipation of energy in the form of heat. This does not happen in superconductors, and as a result a current set up in a circuit made out of only superconducting components can continue to flow indefinitely.

Superconducting states are generally stable at very small temperatures, and a search for a high temperature superconductor is one of the holy grails of condensed matter physics.

Initially, the properties of superconductors were described by Landau-Ginzburg Theory, but this did not incorporate a microscopic justification for the phenomena. This was later achieved in the BCS Theory (Bardeen-Cooper-Schrieffer) in 1957, which proposed that the effect was due to the formation of electron pairs, known as Cooper pairs, which are held together by electron-phonon interactions.

6.1.1. Flux Quantization

According to classical electromagnetism, changing electric fields (or equivalently, currents) create magnetic fields and changing magnetic flux creates electric fields. Classically, magnetic flux can take on a continuum of values. But in a quantum mechanical treatment of electromagnetism, magnetic flux can only take on discrete, quantized values.

In 1961, two independent experimental groups created a cylindrical superconductor and applied a magnetic field along the axis of the cylinder. They then measured the magnetic flux through the cylinder, and found that it was quantized. The following features became apparent from the data:

- The flux was quantized into discrete steps. In contrast, the magnetic response of a normal metal gives us a linear relationship.
- The steps in flux were the same, upto an error of 1%.
- The value of each step was equal to integer multiples of $\frac{h}{2e}$ - a constant formed entirely out of microscopic fundamental constants.

Similar quantization was observed in different materials in their superconducting states, and it also shows up as vortices in the case of Type-II superconductors. This phenomena was the first macroscopic quantum phenomena to be observed.

6.1.2. The Josephson Effect

A Josephson Junction is a device that consists of two superconductors separated by a thin insulating barrier. In 1962, Brian Josephson predicted that such a device would allow for a current to flow across the junction, in the absence of any applied voltage, due to quantum mechanical tunneling of Cooper pairs across the barrier. This current is known as the Josephson current. [3]

Going further, Josephson also predicted that if there was voltage across the junction was varied, the current through it would jump by discrete steps only when the voltage was an integer multiple of $\frac{h}{2e}$. This is called the AC Josephson effect, and it is a remarkably useful macroscopic quantum phenomena in metrology. It is currently used as the standard definition of a Volt, and it can be used to measure the ratio $\frac{h}{e}$ with

incredible accuracy. This was possible as the phenomena was quantum and did not depend on the materials used to construct the junction.

A device that consists of two branching paths of superconducting material, with a Josephson junction at the end of each branch is called a SQUID (Superconducting Quantum Interference Device). As the name suggests, there is a quantum interference effect that can be observed between the currents in the two branches of a SQUID. If the magnetic flux passing through a SQUID device changes, the interference pattern also changes with it (relative to the quantum of magnetic flux). SQUIDs are used to make very precise measurements of magnetic fields, and as qubits in quantum computation.

6.2. Superfluids

Superfluidity was independently discovered in liquid helium (specifically helium-4) in 1937 by two groups: John Allen and Don Misener, and Pyotr Kapitsa. The transition to a superfluid state in helium is marked by a “lambda transition” (at about 2.2 K), a point where the liquid’s specific heat capacity spikes and becomes extremely large. Later, even helium-3 was shown to have a superfluid state at very low temperatures. Bose-Einstein Condensates, which are supercooled gases of bosons are also superfluid.

Superfluids are characterized by their ability to flow without any resistance, i.e. they have zero viscosity. This nicely parallels the flow of electricity without resistance through a superconductor.

For any liquid being rotated at some angular velocity in a container, we can define a quantity called *circulation*. For normal, classical fluids the values taken by circulation are continuous. But for superfluids, circulation is quantized. These discrete steps in circulation were shown to be integer multiples of $\frac{h}{M}$, where M is the mass of the helium atom. This is another macroscopic quantum phenomena.

7. Topological Phases of Matter

Topology is a field of mathematics that studies the properties of shapes that do not change when the shape is *deformed continuously*. This means that there are no cuts or holes poked in the shape in the transformation.

The qualitative differences between different states of matter was originally formalized in terms of symmetry/order. Soon, new states of matter were discovered that could not be adequately described using these concepts alone. These states require the use of topology to understand them.

7.1. Integer Quantum Hall Effect

The Hall effect was first observed by Edwin Hall in 1879. It consists of the production of a voltage difference (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and a magnetic field perpendicular to the current. The Hall effect does not require the use of quantum mechanics to explain it - just considering the electrons as charge carriers with a definite velocity, experiencing a Lorentz force is enough to explain the phenomenon.

After the advent of quantum mechanics, Lev Landau was able to solve for the wavefunction of an electron in a magnetic field. Doing so, he found that the energy levels of the electron were discrete, and that they were equally spaced. These energy levels are called Landau levels. The observation of Landau Levels required very high magnetic fields, pure materials, and low temperatures.

Eventually, it became possible to conduct the Hall effect experiment in very pure, two-dimensional materials at very low temperatures and very high magnetic fields. Klaus von Klitzing was able to carry out this experiment in 1980, and he found that the measured Hall resistance of the sample was quantized. This quantization was not dissimilar to the quantization of magnetic flux and other macroscopic quantum phenomena described before. The steps were quantized to integer multiples of $\frac{h}{2e^2}$, and the quantization was not affected by the material or the charge carriers involved.

The theoretical justification for this effect is that the Hall resistance is a topological invariant. It is not the topology of the real space in which the system resides, but rather the topology of the space that describes the states of the charge carriers in a homogenous magnetic field and a periodic electric field (due to the lattice). The integers here can be thought of as the number of “holes” in the surface that describes the system in this space. It is this connection to topology that leads to the quantum Hall effect being robust to disorder, and other microscopic details.

7.2. Fractional Quantum Hall Effect

Soon after the observation of the integer quantum Hall effect, it became possible to push into a regime beyond the $n = 1$ plateau observed by von Klitzing. In these

regime, more plateaus were discovered but these were not integer multiples of the quantum. Instead, these were observed to be at fractional values, like $\frac{1}{3}$, $\frac{2}{5}$, etc. with odd denominators.

This did not make sense in the framework that explained the Integer quantum Hall effect, but Robert Laughlin was able to provide an explanation for this phenomena by considering the interactions between the electrons in the 2D material. His theory proposed that at each plateau, the electrons collectively act in a way that it appears as if each electron has fallen apart into several quasiparticles that carry a fraction of the electron's charge.

7.3. Quantum Spin Chains

Let us consider the case of the Ising Model again, but now let us allow for both integer and half-integer spins in the system. With such a model in 3 dimensions, there is long-range order (in the Landau sense) of the spins for temperatures that are low enough.

But if we are to consider a 1 dimensional chain of such spins instead, the situation is different. It was shown by Duncan Haldane that there is a fundamental difference between a spin chain with integer spins as compared to spin chains with half-integer spins. For the former, the spins could only be correlated over short distances even at zero temperature, whereas for the latter, there was long range correlation in the spins. Haldane was able to show that this arises from fundamentally topological reasons.

Haldane's theoretical predictions were eventually verified via numerics and experiment, where it was shown that while there was no long-range order for integer spin chains in the Landau sense, there was a sort of *topological order* in the system - it acted as if there were two half-integer spins at the ends of the chain. It was shown in 2010 that this topological order was related to the phenomena of quantum entanglement.

7.4. Topological Insulators

Inspired by the "chiral anomaly" demonstrated by pions in the context of particle physics, Haldane proposed a model where the integer quantum Hall effect could occur without a magnetic field. For this, we studied a model of graphene where the motion of the electrons was different based on whether it was travelling backwards or forwards in time.

Experiments and further theoretical work in this direction led to the discovery of the topological insulator state, where the material acts as an insulator in the bulk, but as a conductor on its edges.

7.5. How topology completes the picture

As can be seen from the various examples provided in the sections above, Landau's formulation of states of matter in terms of symmetry and order was fundamental in-

complete. This was because Landau's characterisation only considered local properties of the material. Topology grants us the ability to consider *global* details of the system as well, and thus it is required for a complete characterisation of the states of matter.

8. The Nature of Emergence

Circling back to the discussion of emergence, last visited in section 2.1., we can now put the idea of it in context of the various systems discussed in the book.

Referring to a scientific discipline as “fundamental”, we often mean to say that the discipline in question is *reductionist*. This means that the discipline seeks to study and understand the underlying physical laws of the universe which gives rise to everything else. What the discipline of Condensed Matter Physics shows us through ample example is that it is a fallacy to believe that such a reduction necessarily implies a constructionist approach. Unlike mass or charge, not all physical properties are *additive*.

8.1. Anderson’s conception of broken symmetry

In his famous article, “More is Different” [4], physicist Philip Anderson argued that the study of increasing complex systems necessarily requires a sort of “fundamental” thinking that is not different, qualitatively, from the “fundamental” thinking that is used to study the simplest systems. At each layer of complication (and scale), we require new concepts and ideas to “fit” the underlying laws and justify/predict the behaviour of the systems in question.

Anderson supplements his ideas with a concrete example - that of the ammonia molecule. Normally, chemists are of the view that the Ammonia molecule can be described by a particular pyramidal shape constructed out of atoms. By virtue of this shape, they predict that the ammonia molecule has a finite dipole moment. This holds up in experiments, and is a feature that is crucial to actually doing chemistry with ammonia. But, on the other hand, from the view of the underlying quantum mechanics, no *stationary state* can have a non-zero dipole moment! There is an apparent contradiction here. Via quantum mechanical tunnelling, the nitrogen atom in ammonia can “leak” out of its position at the top of the pyramid and assume a position on the opposite side - effectively inverting the shape. A truly “stationary state” of ammonia is thus a quantum superposition of two unsymmetrical pyramidal states each of which do not respect the underlying symmetry, and it is this state that has no net dipole moment. While ammonia does oscillate between these inversions at a very high frequency, we can construct heavier molecules like trifluorides that tunnel so slowly that, in comparison to the age of the universe, they essentially do have a non-zero dipole moment.

What Anderson is trying to relay through this example is that a complex system generally tries to relax to a “state” where its energy is minimal. By the symmetry of the underlying laws, there may be another state that has the exact same energy and transitions between the two are not forbidden. But the rates of such transitions may scale in such a way that the system appears to settle into one of the two states. So to describe the system *effectively*, we have to resort to language that does not respect the symmetries of the underlying laws. This is Anderson’s idea of “broken symmetry”, and

it underlines the unique sort of thinking that is required to effectively do condensed matter physics.

9. Open Frontiers in Condensed Matter Physics

At the end of the book, Ross goes over some important open questions that define the frontiers of Condensed Matter Physics at the time of the book's publication. Some of these are:

- Room Temperature superconductivity: The search for a material that can display a superconducting state at room temperatures is the holy grail of condensed matter research, as it can revolutionise many technologies.
- Glass and the Glassy transition: Glass is an unique state of matter that is defined by a lack of long-range order in the same way as crystalline solids, yet it is not a liquid. The problem of relating the macroscopic properties of glass to its underlying disorder is a hard one.
- Quantum Computation: The construction of quantum computers require a utilization of macroscopic quantum phenomena, and reliably conjuring the effects in devices such as SQUIDs is an ongoing research problem.
- States in extreme conditions: Exploring the behaviour of matter under extreme conditions such as very high pressures and temperatures in the context of the interior of planets and stars remains an exciting avenue of research.
- Non-equilibrium states: There is a lot of interest in studying states of matter that change with time, in response to some impulse. All the examples referred to in Ross' book are equilibrium states of matter, and non-equilibrium statistical mechanics is a developing field of study.

In addition to these, Ross ends by asserting that Condensed Matter Physics has huge potential for interdisciplinary work in applied technologies, biology, and even fields like ethology where emergent phenomena are of central importance.

Bibliography

- [1] R. H. McKenzie and R. H. McKenzie, *Condensed Matter Physics: A Very Short Introduction*. in Very Short Introductions. Oxford, New York: Oxford University Press, 2023.
- [2] C. Yang *et al.*, “Signatures of a Strange Metal in a Bosonic System,” *Nature*, vol. 601, no. 7892, pp. 205–210, Jan. 2022, doi: 10.1038/s41586-021-04239-y.
- [3] B. D. Josephson, “Possible New Effects in Superconductive Tunnelling,” *Physics Letters*, vol. 1, no. 7, pp. 251–253, July 1962, doi: 10.1016/0031-9163(62)91369-0.
- [4] “More Is Different | Science.” Accessed: June 10, 2026. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.177.4047.393>